

Dalitz Plot Interactive Web Visualizer - Updated Project Plan

1. Reactive architecture: every change updates the model and plots automatically.
2. Decay selection: user selects mother and three daughters. Backend uses particles and qrules to validate automatically and compute kinematics.
3. Suggested resonances: generated automatically from qrules/particles, grouped by daughter pair.
4. Custom resonances: users may add any resonance manually, either from the particle database or as a completely custom state. The application warns about inconsistencies but never blocks them.
5. Amplitude model: modular components (lineshape, spin factor, Blatt-Weisskopf, complex coefficient). Initial lineshapes: RBW, Gounaris-Sakurai, Flatte.
6. Reactive visualization: Dalitz plot, three invariant-mass projections, optional interference and phase maps.
7. Monte Carlo: weighted generation (up to 1M events), CSV export containing four-momenta, invariant masses and weights.
8. Extensibility: future support for K-matrix, LASS, Argand plots, fit fractions, ROOT/Parquet export, model sharing via JSON.

Design principles:

- Suggest, don't restrict.
- Everything is editable.
- Everything is reactive.